

Coding & Solution

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the coding standards, project structure, readability, and reusable design implemented in the **HDI Predictor** project. The application follows a modular architecture using Machine Learning and Flask to predict the Human Development Index (HDI) based on socio-economic indicators. The project emphasizes clean coding practices, maintainability, and reusable components for efficient development.

Instructions:

- The project code is organized into separate modules for data preprocessing, model training, prediction, and Flask application development.
- The application follows Python coding standards (PEP 8) with meaningful naming conventions and proper indentation for better readability.
- HTML templates, CSS stylesheets, Machine Learning models, datasets, and configuration files are organized into separate folders for easy maintenance.
- Reusable functions, configuration settings, and template structures are implemented to improve scalability and reduce code duplication.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/ashishpaulgunipe-1177/A-Comprehensive-Measure-of-Well-Being

Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy

Code Quality Features Implemented

- Modular project architecture
- Data preprocessing and feature engineering
- HDI prediction using Machine Learning model
- Flask-based web application
- Responsive user interface
- Model loading and prediction using serialized files
- Reusable HTML templates and CSS components

Reusability Features

Component	Implementation
HTML Templates	Common layout reused across multiple pages using Flask templates.
CSS Styling	Shared styles for navigation bar, forms, cards, and buttons.
Configuration File	Model paths and dataset locations stored centrally.
Machine Learning Model	Trained model loaded once and reused for every prediction request.

Component	Implementation
Utility Functions	Data preprocessing and prediction logic written as reusable functions.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes